SONGWEN HUANG

Taipei, Taiwan | +886 909 574 243 | kouppie0414@gmail.com | GitHub: [TBD] | Portfolio: [TBD]

Education

M.S., Life Science, National Taiwan University (NTU) Sep 2024 - Aug 2026
- GPA 4.15 | Rank 7/29

B.S., Life Science, National Taiwan University (NTU) Sep 2021 - Jul 2024

Research & Project Experience

Master's Thesis Research | Advisor: Prof. Yu-Wei Wu Sep 2024 - Aug 2026
- **Hybrid 3D Microelectrode Array (H3D-MEA):** Extending my NSTC undergraduate project and B.S. thesis, developed a 3D microelectrode array combining template-patterned microwires with high-density CMOS readout, with customizable electrode position and depth and verified height-offset alignment; implanted in mouse M1 to record local field potentials and identified single units by spike sorting.
- **64-Channel Wireless Brain-Computer Interface:** Ongoing collaboration with BioPro Scientific and iST (Integrated Service Technology); lead overall system architecture integration and electrode fabrication, integrating two different fabrication processes into a wireless system for chronic recording in freely moving mice.
- **Brain Penetration Mechanics with Microscale Probes:** Performed ultra-sensitive measurements of probe insertion mechanics and blood vessel rupture; published in PNAS (2026) as co-first author.

Desktop Quadruped Companion Robot Dec 2025
- Independently built a Raspberry Pi 5 companion robot integrating vision, audio, locomotion, and on-device edge computing; it recognizes users, understands conversation, and responds with voice and facial expressions, with real-time monitoring and control through a web dashboard.

Publications, Patents & Talks

- **Journal Article** Ultra-sensitive measurement of brain penetration mechanics and blood vessel rupture with microscale probes. *PNAS, Mar 2026 (co-first author)*
- **Patent** Methods for Producing Patterned Microwire Arrays and Uses Thereof (U.S. Provisional Application No. 64/040,119)
- **Invited Young Speaker** 2025.06 Suzhou, Neuroscience Technology Symposium, Chinese Neuroscience Society
- **Invited Young Speaker** 2026 CYCU-Leuven Joint International Workshop
- **Conference Presentations** Japan Neuroscience Society Annual Meeting (JNS, 2025) | TSfN Interdisciplinary Neuroscience International Conference (2024) | Academia Sinica NPAS Summer Undergraduate Research Program, 5th Place (2022)

Skills

- **Mechanical Design & Fabrication:** SolidWorks, Shapr3D (parametric modeling, engineering drawings, tolerancing); FDM/DLP 3D printing, manual milling, drilling and tapping, laser cutting
- **Mechatronics & Embedded Systems:** Arduino, ESP32, Raspberry Pi; servo/stepper motor control and drivers, PWM, sensor integration (IMU, vision, audio modules), circuit assembly and hardware debugging; Python (fundamentals), PySide6, HTML/CSS/JavaScript (vibe-coding projects)
- **Microfabrication & Characterization:** microelectrode fabrication, oxygen plasma treatment, wet etching, laser ablation; SEM, EIS, electrical impedance testing and analysis, two-photon microscopy
- **Neuroengineering & Biology:** mouse stereotaxic surgery, craniotomy, electrophysiological recording, animal behavior experiments, behavioral apparatus development, cardiac perfusion, cryosectioning, IHC
- **Languages:** English (TOEIC 835), Mandarin Chinese (native)

Honors & Awards

- **2026** Taiwan Global Pathfinders Initiative | NTU Life Science Dean's Award | NTU Academic Research Excellence Award | Graduation Poster Competition, 1st Place
- **2025** NTU College of Life Science International Academic Exchange Scholarship | Tri-University Alliance Architectural Photography Course, 1st Place

Extracurricular Activities

- President, NTU Transfer Students Association | Vice President, Life Science Student Association | Art & Publicity Officer, NTU Palm Breeze Rock Club | Advisor, Chenggong High School Medical Research Club
- Freelance Photographer: Official photographer for an EMBO international conference and for NTU student clubs and departments